

Ordinary Differential Equations, Dynamical Systems, and Control Theory

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Contents

Chapter 1 Ordinary Differential Equations (常微分方程), Dynamical Systems (動態系統 or 動力系統), and Control Theory (控制理論)

1 Ordinary Differential Equations

I Definitions

i Normal form

The normal form of an n th-order ODE of a dependent variable y in codomain Y with respect to an independent variable x in domain X is

$$\frac{d^n y}{dx^n} = F(x, y, y', y'', \dots, y^{(n-1)}),$$

in which F is a function of $(n + 1)$ variables with codomain Y .

ii System of ODEs (常微分方程組)

A system of ODEs is two or more equations involving the derivatives of two or more unknown functions or dependent variables of a single independent variable.

iii Initial-value problem (IVP) (初值問題)

An n th-order initial-value problem is an n th-order ODE with n conditions for y and all k th derivative functions with $k \in \mathbb{N} \wedge k < n$ of y at the same point $x = x_0$, that is,

$$\{y^{(k)}(x_0) = y_k \mid k \in \mathbb{N} \wedge k < n\},$$

called initial conditions (ICs) (初始條件).

A solution of an IVP is a solution of the ODE that complies with all the ICs.

The interval of existence of an IVP is the largest open interval containing x_0 where a solution of the IVP exists; the interval of existence of an IVP is the largest open interval containing x_0 where a unique solution of the IVP exists.

iv Boundary-value problem (BVP) (邊值問題)

An n th-order boundary-value problem is an n th-order ODE with some conditions called boundary conditions (BCs) (邊界條件).

A solution of an BVP is a solution of the ODE that complies with all the BCs.

II Solutions of ODEs

i Solution or flow of an ODE

Any function f that is defined on an interval I , called interval of definition, the interval of existence, the interval of validity, or the domain of the solution, and of class C^n on I , and when substituted into an n th-order ordinary differential equation reduces the equation to an identity, is said to be a solution (aka flow) of the equation on I .

An ODE does not necessarily have to possess a solution.

ii Solution curve or trajectory

The graph of a solution f on its interval of definition of an ODE is called a solution curve or a trajectory.

iii Explicit solution

An explicit solution of an ODE of a dependent variable y with respect to an independent variable x is in the form $y = f(x)$, where $f(x)$ is a function of x .

iv Implicit solution

A relation $G(x, y) = 0$ is said to be an implicit solution of an ODE on an interval I if there exists at least one function f that satisfies the relation $G(x, y) = 0$ as well as the ODE on I .

v Parametric family or parameterized family

A parametric family or a parameterized family is a family of objects whose differences depend only on the chosen values for a set of parameters.

vi Degrees of freedom

The degrees of freedom of a system is the number of parameters of the system that may vary independently.

vii Parametric family of solutions

An n -parameter family of solutions of an ODE of a dependent variable y with respect to an independent variable x is in the form $y = f(x, c_1, c_2, \dots, c_n)$ (explicit) or $G(x, y, c_1, c_2, \dots, c_n) = 0$ (implicit), in which $c_1, c_2, \dots, c_n$ are parameters that are arbitrary given that the solution obtained is a solution of the ODE. The arbitrary parameters are usually denoted as $A, B, \dots$.

viii General solution (通解)

If every solution of an ODE on an interval I can be obtained from an n -parameter family of solutions by appropriate choices of the parameters, we then say that that family of solutions

is the general solution of the ODE.

ix Particular solution (特解)

A particular solution of an ODE that is free of parameters is called a particular solution.

If a system given by an ODE have n degrees of freedom, one may obtain a determined particular solution by imposing $n - 1$ non contradictory constraints, leaving one degree of freedom determined by the ODE.

x Singular solution (奇異解)

Sometimes a differential equation possesses a solution that is not a member of a family of solutions of the equation, that is, a solution that cannot be obtained by specializing any of the parameters in the family of solutions. Such an extra solution is called a singular solution.

xi Solutions by substitutions or solutions by change of variables

When solving an ODE, we often replace dependent variables with functions of the independent variable and new dependent variables where the new dependent variables are functions of the independent variable and old dependent variables to transform it into another ODE of the new dependent variables that is easier to solve.

xii Solution of System of ODEs

A solution of a system of ODEs involving the derivatives of n unknown functions or dependent variables, $y_1, y_2, \dots, y_n$ of a single independent variable x is a n -tuple of sufficiently smooth functions of x defined on a common interval of definition that satisfy all equations in the system.

A system of ODEs does not necessarily have to possess a solution.

xiii Orthogonal trajectory and isogonal trajectory

An orthogonal trajectory of a family of curves is a curve that intersects each curve of the family orthogonally.

An isogonal trajectory of a family of curves is a curve that intersects each curve of the family at a same angle α .

III Conversion of order to variable

Any n th-order system of ODEs of m variables $x_1, x_2, \dots, x_m$ can be converted to a first-order system of ODEs of $m \cdot n$ variables by, for each variable x , assigning $(n - 1)$ new variables defined as the first to $(n - 1)$ th derivatives of x .

2 First-order ODEs

Unless otherwise specified, the variables and functions are in $\mathbb{R}$.

I Differential form

The differential form of a first-order ODE $M(x, y) + N(x, y) \frac{dy}{dx} = 0$ is

$$M(x, y) dx + N(x, y) dy = 0.$$

II Slope

i Slope function or rate function

In a first-order ODE in the form $\frac{dy}{dx} = f(x, y)$, f is called the slope function or rate function.

ii Direction field or slope field (斜率場)

In a first-order ODE in the form $\frac{dy}{dx} = f(x, y)$, if we evaluate f over a rectangle grid of points and draw a lineal element at each point (x, y) of the grid with slope $f(x, y)$, then the collection of all these lineal elements is called a direction field or a slope field of the ODE.

iii Isocline

In a first-order ODE in the form $\frac{dy}{dx} = f(x, y)$, the isocline for slope m is defined as the family of points $\{(x, y) \mid f(x, y) = m\}$. In the method of isocline, we draw a lineal element at each point in the isocline for slope m of f with slope m .

iv Euler method or forward Euler method

Approximate values for the solution of the initial-value problem $y' = f(x, y)$, $y(x_0) = y_0$, with step size h , at $x_n = x_{n-1} + h$, are

$$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1}), \quad n \in \mathbb{N}.$$

v Backward Euler method

Approximate values for the solution of the initial-value problem $y' = f(x, y)$, $y(x_0) = y_0$, with step size h , at $x_n = x_{n-1} + h$, are the solutions of the equations

$$y_n = y_{n-1} + hf(x_n, y_n), \quad n \in \mathbb{N}.$$

III Autonomous System

i Autonomous system (自治系統)

A system of n th-order ODEs is called autonomous if it can be written in the form

$$\frac{d^n x}{dt^n} = f\left(x, \frac{dx}{dt}, \dots, \frac{d^{n-1}x}{dt^{n-1}}\right).$$

ii Critical point (臨界點), equilibrium, equilibrium point (平衡點), or stationary point (駐點) of autonomous systems of ODEs

For an autonomous systems of ODEs, we say that a point c in the domain of f is a critical point, equilibrium, equilibrium point, or stationary point if $f(c) = 0$.

If c is a critical point, then $y(x) = c$ is a constant solution of the system of autonomous first-order ODEs, also called an equilibrium solution (平衡解).

If c is a critical point and f is locally Lipschitz at c , then $y(x) = c$ is the unique solution of the system of autonomous first-order ODEs through c .

iii Translation property or shift property of solutions of autonomous systems of ODEs

Let function $\Phi(t, y)$ denotes the solution of an autonomous system of ODEs with IC $x(0) = y$. The translation property or shift property states that

$$\forall s \in \mathbb{R} : \Phi(t + s, y) = \Phi(t, \Phi(s, y)).$$

IV Orthogonal trajectories

The family of the orthogonal trajectories of the family of solutions of the first-order ODE

$$M(x, y) dx + N(x, y) dy = 0,$$

is the family of solutions of the first-order ODE

$$N(x, y) dx - M(x, y) dy = 0.$$

V Separable equation

i Definition

A first-order ODE in the form

$$\frac{dy}{dx} = g(x)h(y)$$

is said to be separable or have separable variables.

ii Solutions by direct integration

The solutions of

$$\frac{dy}{dx} = g(x)h(y)$$

are the family of solutions

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

and the constant solution

$$h(y) = 0.$$

iii Example: law of natural growth

IVP:

$$\frac{dP}{dt} = kP, \quad P(0) = P_0.$$

Solutions:

$$P = P_0 e^{kt}.$$

Proof.

$$\begin{aligned} P^{-1} dP &= k dt \\ \int P^{-1} dP &= \int k dt = kt + A = \ln(P). \end{aligned}$$

$$P = A e^{kt}.$$

$$P_0 = A.$$

□

iv Example: logistic differential equation

IVP:

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{M}\right), \quad P(0) = P_0.$$

Solutions:

$$P = \frac{M}{1 + \frac{M-P_0}{P_0} e^{-kt}}.$$

Proof.

$$\begin{aligned} \frac{1}{P \left(1 - \frac{P}{M}\right)} dP &= k dt \\ \int \frac{1}{P \left(1 - \frac{P}{M}\right)} dP &= \int k dt = kt + A \\ \frac{1}{P \left(1 - \frac{P}{M}\right)} &= \frac{M}{P(M-P)} = \frac{A}{P} + \frac{B}{M-P} \\ AM - AP + BP &= M, \quad A = 1, \quad B = 1 \\ \int \frac{1}{P \left(1 - \frac{P}{M}\right)} dP &= \int \frac{1}{P} + \frac{1}{M-P} dP = \ln |P| - \ln |M-P| + C \\ \ln |P| - \ln |M-P| &= kt + A = \ln \left| \frac{P}{M-P} \right| \\ \frac{P}{M-P} &= A e^{kt} \\ P &= \frac{AM e^{kt}}{1 + A e^{kt}} \\ P_0 &= \frac{AM}{1 + A} \\ A &= \frac{P_0}{M - P_0} \end{aligned}$$

$$P = \frac{\frac{P_0}{M-P_0} M e^{kt}}{1 + \frac{P_0}{M-P_0} e^{kt}} = \frac{M}{1 + \frac{M-P_0}{P_0} e^{-kt}}$$

□

v Example: Lotka–Volterra equations or predator–prey equations

ODE:

$$\begin{aligned}\frac{dx}{dt} &= x(\alpha - \beta y), \\ \frac{dy}{dt} &= -y(\gamma - \delta x).\end{aligned}$$

Implicit solutions:

$$\delta x - \gamma \ln(x) + \beta y - \alpha \ln(y) = C.$$

Equilibrium solutions:

$$x = 0, \quad y = 0,$$

and

$$x = \frac{\gamma}{\delta}, \quad y = \frac{\alpha}{\beta}.$$

Proof.

$$\begin{aligned}\frac{dy}{dt} &= \frac{dy}{dx} \frac{dx}{dt} \\ \frac{dy}{dx} &= -\frac{y}{x} \frac{\gamma - \delta x}{\alpha - \beta y} \\ \frac{\delta x - \gamma}{x} dx &= -\frac{\beta y - \alpha}{y} dy \\ \delta x - \gamma \ln(x) + \beta y - \alpha \ln(y) &= C\end{aligned}$$

□

VI Linear equation

i Definition

A first-order ODE in the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

is said to be linear.

ii Solutions by integrating factor

Use the integrating factor:

$$\mu(x) = e^{\int P(x) dx}.$$

Multiply both sides with $\mu(x)$:

$$\mu(x) \frac{dy}{dx} + \mu(x) P(x)y = \mu(x) Q(x).$$

The left-hand side is exactly a derivative:

$$\frac{d}{dx} (\mu(x)y(x)) = \mu(x)Q(x).$$

Integrate:

$$\begin{aligned}\mu(x)y(x) &= \int \mu(x)Q(x) dx + C. \\ y(x) &= \frac{1}{\mu(x)} \left(\int \mu(x)Q(x) dx + C \right).\end{aligned}$$

iii Remarks

Occasionally, a first-order differential equation is not linear in one variable but is linear in the other variable. If a first-order ODE that can be written in the form

$$\frac{dy}{dx} = \frac{1}{P(y)x + Q(y)}.$$

We can take its reciprocal as

$$\frac{dx}{dy} - P(y)x = Q(y),$$

which is recognized as linear in the variable x .

iv Example: constant voltage series RL filter circuit

IVP:

$$L \frac{dI(t)}{dt} + RI(t) = V, \quad I(0) = I_0.$$

Solutions:

$$I(t) = \frac{V}{R} (1 - e^{-Rt/L}) + I_0 e^{-Rt/L}.$$

Equilibrium solution:

$$I(t) = \frac{V}{R}.$$

Proof.

$$\begin{aligned}\frac{dI(t)}{dt} + \frac{R}{L}I(t) &= \frac{V}{L} \\ \mu(t) &= \exp\left(\int \frac{R}{L} dt\right) = e^{Rt/L} \\ e^{Rt/L} \frac{dI(t)}{dt} + e^{Rt/L} \frac{R}{L}I(t) &= \frac{du}{dt}, \quad u = e^{Rt/L} I(t) \\ \int 1 du &= \int e^{Rt/L} \frac{V}{L} dt = u + A = \frac{V}{R} e^{Rt/L} + B \\ u &= e^{Rt/L} I(t) = \frac{V}{R} e^{Rt/L} + A \\ I(t) &= \frac{V}{R} + A e^{-Rt/L} \\ I_0 &= \frac{V}{R} + A \\ A &= I_0 - \frac{V}{R} \\ I(t) &= \frac{V}{R} (1 - e^{-Rt/L}) + I_0 e^{-Rt/L}\end{aligned}$$

□

v Example: constant current parallel RC filter circuit

IVP:

$$C \frac{dV(t)}{dt} + \frac{1}{R} V(t) = I, \quad V(0) = V_0.$$

Solutions:

$$V(t) = IR \left(1 - e^{-t/(RC)}\right) + V_0 e^{-t/(RC)}.$$

Equilibrium solution:

$$V(t) = IR.$$

Proof.

$$\begin{aligned} \frac{dV(t)}{dt} + \frac{1}{RC} V(t) &= \frac{I}{C} \\ \mu(t) &= \exp\left(\int \frac{1}{RC} dt\right) = e^{t/(RC)} \\ e^{t/(RC)} \frac{dV(t)}{dt} + e^{t/(RC)} \frac{1}{RC} V(t) &= e^{t/(RC)} \frac{I}{C} = \frac{du}{dt}, \quad u = e^{t/(RC)} V(t) \\ \int 1 du &= \int e^{t/(RC)} \frac{I}{C} dt = u + A = IR e^{t/(RC)} + B \\ u &= e^{t/(RC)} V(t) = IR e^{t/(RC)} + A \\ V(t) &= IR + A e^{-t/(RC)} \\ V_0 &= IR + A \\ A &= V_0 - IR \\ V(t) &= IR \left(1 - e^{-t/(RC)}\right) + V_0 e^{-t/(RC)} \end{aligned}$$

□

vi Example: Falling with drag force complying with Stokes' law

IVP:

$$m \frac{dv}{dt} = -kv - mg, \quad v(0) = v_0.$$

Solutions:

$$v = \frac{mg}{k} \left(1 - e^{-kt/m}\right) + v_0 e^{-kt/m}.$$

Equilibrium solution:

$$v = \frac{mg}{k}.$$

Proof.

$$\begin{aligned} \frac{dv}{dt} + \frac{k}{m} v &= -g \\ \mu(t) &= \exp\left(\int \frac{k}{m} dt\right) = e^{kt/m} \\ \frac{d}{dt} (e^{kt/m} v) &= -e^{kt/m} g \\ e^{kt/m} v &= \int -e^{kt/m} g dt = \frac{mg}{k} e^{kt/m} + C \end{aligned}$$

$$v = \frac{mg}{k} + Ce^{-kt/m}$$

$$v_0 = \frac{mg}{k} + C$$

$$v = \frac{mg}{k} + \left(v_0 - \frac{mg}{k}\right) e^{-kt/m}.$$

□

VII Exact equation

i Definition

A first-order ODE in the form

$$M(x, y) dx + N(x, y) dy = 0$$

where $\frac{\partial^2 M}{\partial x \partial y}$ and $\frac{\partial^2 N}{\partial x \partial y}$ is continuous and

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

is said to be exact.

By Schwarz's Theorem, the criterion holds iff there exists a function $F : R \rightarrow \mathbb{R}$ such that

$$\frac{\partial F}{\partial x} = M, \quad \frac{\partial F}{\partial y} = N,$$

and $\frac{\partial^2 F}{\partial x \partial y}$ is continuous. Such F is called the potential function.

ii Solutions by integration and then differentiation with respect to different variables

First, integrate $M(x, y)$ with respect to x :

$$F(x, y) = \int M(x, y) dx + \phi(y),$$

where $\phi(y)$ is an arbitrary constant of integration that may depend on y .

Second, differentiate with respect to y :

$$\frac{\partial F}{\partial y}(x, y) = N(x, y) = \frac{\partial}{\partial y} \left(\int M(x, y) dx \right) + \phi'(y).$$

Third, integrate $\phi'(y)$ with respect to y to find $\phi(y)$:

$$\phi(y) = \int \phi'(y) dy = \int N(x, y) - \frac{\partial}{\partial y} \left(\int M(x, y) dx \right) dy.$$

Fourth, substitute $\phi(y)$ back to find the potential function $F(x, y)$. The implicit solutions is given by

$$F(x, y) = C,$$

where C is an arbitrary constant.

Alternatively, first, integrate $N(x, y)$ with respect to y :

$$F(x, y) = \int N(x, y) dy + \phi(x),$$

where $\phi(x)$ is an arbitrary constant of integration that may depend on x .

Second, differentiate with respect to x :

$$\frac{\partial F}{\partial x}(x, y) = M(x, y) = \frac{\partial}{\partial x} \left(\int N(x, y) dy \right) + \phi'(x).$$

Third, integrate $\phi(x)$ with respect to x to find $\phi(x)$:

$$\phi(x) = \int \phi'(x) dx = \int M(x, y) - \frac{\partial}{\partial x} \left(\int N(x, y) dy \right) dx.$$

Fourth, substitute $\phi(x)$ back to find the potential function $F(x, y)$. The implicit solutions is given by

$$F(x, y) = C,$$

where C is an arbitrary constant.

iii Example: $(4x^3 + 3y^2 + \cos(x)) dx + (6xy + 2) dy = 0$

ODE:

$$(4x^3 + 3y^2 + \cos(x)) dx + (6xy + 2) dy = 0.$$

Solutions:

$$x^4 + 3xy^2 + \sin(x) + 2y = C.$$

Proof.

$$\frac{\partial}{\partial y}(4x^3 + 3y^2 + \cos(x)) = 6y$$

$$\frac{\partial}{\partial x}(6xy + 2) = 6y$$

$$F(x, y) = \int 4x^3 + 3y^2 + \cos(x) dx + \phi(y)$$

$$= x^4 + 3xy^2 + \sin(x) + \phi(y)$$

$$\frac{\partial F}{\partial y} = 6xy + \phi'(y) = 6xy + 2$$

$$\phi'(y) = 2, \quad \phi(y) = 2y + C$$

$$F(x, y) = x^4 + 3xy^2 + \sin(x) + 2y + C$$

□

VIII Integration factors to make inexact equations exact

i General case

Given a first-order ODE in the form

$$M(x, y) dx + N(x, y) dy = 0,$$

we define the integration factor $\mu(x, y)$ such that

$$\mu(x, y)M(x, y) dx + \mu(x, y)N(x, y) dy = 0$$

is exact and with potential function F such that

$$\frac{\partial F}{\partial x} = \mu(x, y)M(x, y), \quad \frac{\partial F}{\partial y} = \mu(x, y)N(x, y),$$

that is,

$$\frac{\partial \mu M}{\partial y} = \frac{\partial \mu N}{\partial x}.$$

Solve the PDE to get μ .

ii Separable case

If

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$$

depends only on x , we can construct integration factor $\mu(x)$:

$$\mu(x) = e^{\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} dx}$$

such that

$$\frac{\partial \mu M}{\partial y} = \frac{\partial \mu N}{\partial x} = \mu \frac{\partial M}{\partial y}.$$

Similarly, if

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

depends only on y , we can construct integration factor $\mu(y)$:

$$\mu(y) = e^{\int \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} dy}$$

such that

$$\frac{\partial \mu M}{\partial y} = \frac{\partial \mu N}{\partial x} = \mu \frac{\partial N}{\partial x}.$$

Proof.

$$\frac{\mu'(x)}{\mu(x)} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}.$$

This is a separable ODE of μ with respect to x .

Let

$$P(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}.$$

$$\mu^{-1} d\mu = P(x) dx.$$

$$\ln|\mu| = \int P(x) dx.$$

$$\mu = e^{\int P(x) dx}.$$

Similarly,

$$\frac{\mu'(y)}{\mu(y)} = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

for the second case. □

IX Example: $(5xy + 4y^2 + 1) dx + (x^2 + 2xy) dy = 0$

ODE:

$$(5xy + 4y^2 + 1) dx + (x^2 + 2xy) dy = 0.$$

Solutions:

$$x^5 y + x^4 y^2 + \frac{x^4}{4} = C.$$

Proof.

$$\frac{\partial}{\partial y}(5xy + 4y^2 + 1) = 5x + 8y$$

$$\frac{\partial}{\partial x}(x^2 + 2xy) = 2x + 2y$$

$$\frac{\frac{\partial}{\partial y}(5xy + 4y^2 + 1) - \frac{\partial}{\partial x}(x^2 + 2xy)}{x^2 + 2xy} = \frac{3x + 6y}{x^2 + 2xy} = \frac{3}{x}$$

$$\mu(x) = \exp\left(\int \frac{3}{x} dx\right) = x^3$$

$$x^3(5xy + 4y^2 + 1) dx + x^3(x^2 + 2xy) dy = 0$$

$$F(x, y) = \int x^3(5xy + 4y^2 + 1) dx + \phi(y)$$

$$= \int 5x^4 y + 4x^3 y^2 + x^3 dx + \phi(y)$$

$$= x^5 y + x^4 y^2 + \frac{x^4}{4} + \phi(y)$$

$$\frac{\partial F}{\partial y} = x^5 + 2x^4 y + \phi'(y) = x^3(x^2 + 2xy) = x^5 + 2x^4 y$$

$$\phi'(y) = 0, \quad \phi(y) = C$$

$$F(x, y) = x^5 y + x^4 y^2 + \frac{x^4}{4} + C$$

□

i Example: $(2xy + y) dx + x^2 dy = 0$

ODE:

$$(2xy + y) dx + x^2 dy = 0.$$

Solutions:

$$y = Ce^{1/x} x^{-2}.$$

Proof.

$$\frac{\partial}{\partial y}(2xy + y) = 2x + 1$$

$$\frac{\partial}{\partial x}(x^2) = 2x$$

$$\frac{\frac{\partial}{\partial y}(2xy + y) - \frac{\partial}{\partial x}(x^2)}{x^2} = \frac{1}{x^2}$$

$$\mu(x) = \exp\left(\int \frac{1}{x^2} dx\right) = e^{-1/x}$$

$$e^{-1/x}(2xy + y) dx + e^{-1/x}x^2 dy = 0.$$

$$F(x, y) = \int e^{-1/x}x^2 dy + \phi(x) = e^{-1/x}x^2y + \phi(x)$$

$$\frac{\partial F}{\partial x} = (x^{-2}e^{-1/x}x^2 + e^{-1/x}2x)y + \phi'(x) = e^{-1/x}(2x + 1)y + \phi'(x)$$

$$\phi'(x) = 0$$

$$F(x, y) = e^{-1/x}x^2y + C$$

$$e^{-1/x}x^2y = C$$

$$y = Ce^{1/x}x^{-2}$$

□

X Homogeneous equation

i Definition

A first-order ODE in the form

$$\frac{dy}{dx} = F(x, y)$$

with $F(x, y)$ be a homogeneous function of degree 0 is said to be homogeneous.

For first-order ODE in the form

$$M(x, y) dx + N(x, y) dy = 0,$$

if $M(x, y)$ and $N(x, y)$ are homogeneous functions of the same degree, $F(x, y) = -\frac{M(x, y)}{N(x, y)}$ is a homogeneous function of degree 0.

ii Solutions by substitutions

For any homogeneous function $F(x, y)$ of 2 variables, there exists a function f of 1 variable such that $F(x, y) = f\left(\frac{y}{x}\right)$ for all $x \neq 0$ such that (x, y) is in the domain of F .

Introduce the substitution $v = \frac{y}{x}$.

$$\frac{dy}{dx} = \frac{d(xv)}{dx} = v + x \frac{dv}{dx} = f(v),$$

which is a separable ODE.

$$\frac{dv}{f(v) - v} = \frac{dx}{x}.$$

$$\int \frac{dv}{f(v) - v} = \ln|x| + C.$$

Substituting $v = \frac{y}{x}$ back gives the solutions.

If F is defined when $x = 0$, we also have a implicit constant solution

$$F(0, y) = 0.$$

iii Example: $(x^2 + y^2) dx - 2xy dy = 0$

ODE:

$$(x^2 + y^2) dx - 2xy dy = 0.$$

Implicit solutions:

$$y^2 = x^2 + Cx.$$

Proof.

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy} = F(x, y)$$

$$F(tx, ty) = F(x, y)$$

$$v = \frac{y}{x}$$

$$F(x, y) = f(v) = \frac{1 + v^2}{2v}$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx} = f(v)$$

$$\frac{1}{f(v) - v} dv = \frac{dx}{x}$$

$$\int \frac{1}{f(v) - v} dv = \ln|x| + C$$

$$= \int \frac{2v}{1 - v^2} dv$$

$$= -\ln|u|, \quad u = 1 - v^2$$

$$= -\ln|1 - v^2|$$

$$C|x| = \frac{1}{|1 - v^2|}$$

$$v^2 = 1 + \frac{C}{x}$$

$$y^2 = x^2 + Cx$$

□

iv Example: $(x^2 + y^2) dx + y^2 dy = 0$

ODE:

$$(x^2 + y^2) dx + y^2 dy = 0.$$

Implicit solutions:

$$\arctan\left(\frac{2\sqrt{3}}{3}\left(\frac{y}{x} + \frac{1}{2}\right)\right) = \frac{\sqrt{3}}{2} \ln\left|\frac{x^2}{y}\right| + C.$$

Constant solution:

$$x = 0, \quad y = 0.$$

Proof.

$$\frac{dy}{dx} = -\frac{y^2}{x^2 + y^2} = F(x, y)$$

$$F(tx, ty) = F(x, y)$$

$$v = \frac{y}{x}$$

$$F(x, y) = f(v) = -\frac{v^2}{1 + v^2}, \quad x \neq 0$$

$$\frac{1}{f(v) - v} dv = \frac{dx}{x}$$

$$\int \frac{1}{f(v) - v} dv = \ln|x| + C$$

$$\frac{1}{f(v) - v} = -\frac{1 + v^2}{v(1 + v + v^2)} = -\frac{A}{v} - \frac{Bv + C}{1 + v + v^2}$$

$$A + Av + Av^2 + Bv^2 + Cv = 1 + v^2$$

$$A = 1, \quad B = 0, \quad C = -1$$

$$\begin{aligned} \int \frac{1}{f(v) - v} dv &= -\int \frac{1}{v} dv + \int \frac{1}{1 + v + v^2} dv \\ &= \ln|v| + \int \frac{1}{\left(v + \frac{1}{2}\right)^2 + \frac{3}{4}} dv, \quad u = \frac{2}{\sqrt{3}}\left(v + \frac{1}{2}\right) \end{aligned}$$

$$= \ln|v| + \frac{\sqrt{3}}{2} \int \frac{1}{u^2 + 1} du$$

$$= \ln|v| + \frac{2\sqrt{3}}{3} \arctan(u) + C$$

$$= \ln|v| + \frac{2\sqrt{3}}{3} \arctan\left(\frac{2\sqrt{3}}{3}\left(v + \frac{1}{2}\right)\right) + C$$

$$\ln|v| + \frac{2\sqrt{3}}{3} \arctan\left(\frac{2\sqrt{3}}{3}\left(v + \frac{1}{2}\right)\right) = \ln|x| + C$$

$$\ln|y| - \ln|x| + \frac{2\sqrt{3}}{3} \arctan\left(\frac{2\sqrt{3}}{3} \left(\frac{y}{x} + \frac{1}{2}\right)\right) = \ln|x| + C$$

$$\arctan\left(\frac{2\sqrt{3}}{3} \left(\frac{y}{x} + \frac{1}{2}\right)\right) = \frac{\sqrt{3}}{2} \ln\left|\frac{x^2}{y}\right| + C$$

$$\arctan\left(\frac{2\sqrt{3}}{3} \left(\frac{y}{x} + \frac{1}{2}\right)\right) = \frac{\sqrt{3}}{2} \ln\left|\frac{x^2}{y}\right| + C$$

$$F(0, y) = 0, \quad y = 0$$

□

XI Bernoulli differential equation

i Definition

A first-order ODE in the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \quad n \in \mathbb{R} \setminus \{0, 1\}$$

is called a Bernoulli differential equation.

Remark: When $n = 0$ or 1 , it is a first-order linear equation.

ii Solutions by substitutions

Divide both sides by y^n :

$$y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x).$$

Introduce the substitution $v = y^{1-n}$.

$$\frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx}.$$

Substitute into the equation:

$$\frac{1}{1-n} \frac{dv}{dx} + P(x)v = Q(x).$$

$$\frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x),$$

which is linear.

Use the integrating factor:

$$\mu(x) = e^{\int (1-n)P(x)dx}.$$

Multiply both sides with $\mu(x)$:

$$\mu(x) \frac{dv}{dx} + \mu(x)(1-n)P(x)v = \mu(x)(1-n)Q(x).$$

The left-hand side is exactly a derivative:

$$\frac{d}{dx} (\mu(x)v(x)) = \mu(x)(1-n)Q(x).$$

Integrate:

$$\mu(x)v(x) = \int \mu(x)(1-n)Q(x) dx + C.$$
$$v(x) = \frac{1}{\mu(x)} \left(\int \mu(x)(1-n)Q(x) dx + C \right).$$

Recall $v = y^{1-n}$, we obtain the family of general solutions.

iii Example: $\frac{dy}{dx} + \cos(x)y = \sin(2x)y^{3/2}$

ODE:

$$\frac{dy}{dx} + \cos(x)y = \sin(2x)y^{3/2}.$$

Solutions:

$$y = \left(2 \sin(x) + 4 + Ce^{\sin(x)/2} \right)^{-2}.$$

Proof.

$$y^{-3/2} \frac{dy}{dx} + \cos(x)y^{-1/2} = 2 \sin(x) \cos(x)$$

$$v = y^{-1/2}, \quad \frac{dv}{dx} = -\frac{1}{2}y^{-3/2} \frac{dy}{dx}.$$

$$\frac{dv}{dx} - \frac{1}{2} \cos(x)v = -\sin(x) \cos(x)$$

$$\mu(x) = \exp\left(\int -\frac{1}{2} \cos(x) dx\right) = e^{-\sin(x)/2}$$

$$\frac{d}{dx} \left(e^{-\sin(x)/2} v \right) = -\sin(x) \cos(x) e^{-\sin(x)/2}$$

$$e^{-\sin(x)/2} v = \int -\sin(x) \cos(x) e^{-\sin(x)/2} dx, \quad u = \frac{\sin(x)}{2}$$

$$= -4 \int u e^{-u} du$$

$$= -4 \left(-u e^{-u} - \int -e^{-u} du \right)$$

$$= -4 \left(-u e^{-u} - e^{-u} + C \right)$$

$$= 2 \sin(x) e^{-\sin(x)/2} + 4 e^{-\sin(x)/2} + C$$

$$v = 2 \sin(x) + 4 + C e^{\sin(x)/2}$$

$$y = v^{-2} = \left(2 \sin(x) + 4 + C e^{\sin(x)/2} \right)^{-2}$$

□

XII Slope being function of linear function

ODE:

$$\frac{dy}{dx} = f(u), \quad u = ax + by + c.$$

Solutions:

$$\int \frac{1}{a + bf(u)} du = x + C.$$

Proof.

$$\frac{du}{dx} = a + b \frac{dy}{dx} = a + bf(u)$$

This is separable.

$$\int \frac{1}{a + bf(u)} du = x + C$$

□

XIII Slope being function of linear function over linear function

ODE:

$$\frac{dy}{dx} = F\left(\frac{ax + by + c}{px + qy + r}\right), \quad aq - bp \neq 0.$$

We can find a unique solution of $ax + by + c = px + qy + r = 0$ for x and y , let it be $x = \alpha$, $y = \beta$, let

$$f(t) = F\left(\frac{a + bt}{p + qt}\right),$$

and we can rewrite it as:

$$\frac{dy}{dx} = f(u), \quad u = \frac{y - \beta}{x - \alpha}.$$

Proof.

$$y = u(x - \alpha) + \beta$$

$$ax + by + c = (a + bu)(x - \alpha) + a\alpha + b\beta + c = (a + bu)(x - \alpha)$$

$$px + qy + r = (p + qu)(x - \alpha) + p\alpha + q\beta + r = (p + qu)(x - \alpha)$$

$$\frac{ax + by + c}{px + qy + r} = \frac{a + bu}{p + qu}$$

□

Solutions:

$$\int \frac{1}{f(u) - u} du = \ln|x - \alpha| + C,$$

and possible constant or singular solutions:

$$f(u) = u.$$

Proof.

$$y = (x - \alpha)u + \beta$$

$$\frac{dy}{dx} = \frac{\partial}{\partial x} ((x - \alpha)u + \beta) + \frac{\partial}{\partial u} ((x - \alpha)u + \beta) \frac{du}{dx} = u + (x - \alpha) \frac{du}{dx}$$

$$u + (x - \alpha) \frac{du}{dx} = f(u)$$

$$\frac{1}{f(u) - u} du = \frac{1}{x - \alpha} dx$$

This is separable.

$$\int \frac{1}{f(u) - u} du = \ln|x - \alpha| + C$$

□

3 Linear ODE

Unless otherwise specified, the variables and functions are in $\mathbb{R}$.

I Definition

An n th-order ODE of a dependent variable y with respect to an independent variable x is linear iff it can be written in the form

$$\sum_{i=0}^n p_i(x)y^{(i)} = q(x),$$

where $p_0(x), \dots, p_n(x)$, called the coefficients, and $q(x)$, called the forcing term, are functions of x , and $p_n(x)$ is nontrivial.

An ODE is nonlinear if it is not linear.

II Linearity

Let $L[y] = \sum_{i=0}^n p_i(x)y^{(i)}$ denotes the left-hand side, then $L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2]$. That's why it's called to be linear.

III Homogeneity

A linear ODE is called a linear homogeneous ODE iff the forcing term is 0 for all x ; otherwise, it is called a linear nonhomogeneous ODE.

IV Superposition principle

If $y_1(x)$ and $y_2(x)$ are both solutions of a linear homogeneous ODE on an interval, then $c_1 y_1(x) + c_2 y_2(x)$ is also a solution of the ODE on the interval for any constants c_1, c_2 .

V Linear independence

n functions of the same domain and codomain $y_1(x), y_2(x), \dots, y_n(x)$ are said to be linearly independent iff

$$\sum_{i=1}^n c_i y_i(x) = 0 \implies \forall i : c_i = 0.$$

VI Solutions of linear homogenous ODE as vector space

The general solutions of an n th-order linear homogenous ODE is a vector space of dimension n .

VII Basis of solutions of linear homogenous ODE

If $y_1(x), y_2(x), \dots, y_n(x)$ are n linearly independent solutions of an n th-order linear homogeneous ODE on an interval, then they are a basis of the vector space of the general solutions of the ODE on the interval, and $\sum_{i=1}^n c_i y_i(x)$ is the family general solutions of the ODE on the interval with n parameter $c_1, c_2, \dots, c_n$.

VIII Complementary equation and particularly solution

Given an n th-order linear nonhomogeneous ODE

$$y^{(n)} + \sum_{i=0}^{n-1} p_i(x)y^{(i)} = q(x),$$

the n th-order linear homogeneous ODE,

$$y^{(n)} + \sum_{i=0}^{n-1} p_i(x)y^{(i)} = 0,$$

is called its complementary equation, and the general solutions of the original equation is the sum of the general solutions y_c of the complementary equation, also called natural solution or homogeneous solution, and a particular solution y_p .

IX Solutions of linear nonhomogenous ODE as affine space

The general solutions of a linear nonhomogenous ODE is an affine space whose associated vector space is the general solutions of its complementary equation.

X Characteristic equation or auxiliary equation for constant coefficient linear homogeneous ODE

i Method

Consider an n th-order constant coefficient linear homogeneous ODE

$$\sum_{j=0}^n a_j y^{(j)} = 0, \quad a_n \neq 0.$$

The formation of its general solutions using the characteristic equation (also called auxiliary equation) is as follows:

1. The characteristic equation is

$$\sum_{j=0}^n a_j r^j = 0.$$

2. Suppose there are k distinct real roots $r_1, r_2, \dots, r_k$, each r_j with multiplicity m_j , and h distinct pairs of conjugate imaginary roots $\alpha_1 \pm \beta_1 i, \alpha_2 \pm \beta_2 i, \dots, \alpha_h \pm \beta_h i$ ($\beta_j \neq 0$), each $\alpha_j \pm \beta_j i$

with multiplicity ω_j . (Thus, $\sum_{j=1}^k m_j + 2 \sum_{j=1}^h \omega_j = n$.)

3. The n -parameter family of general solutions of the ODE is

$$\sum_{j=1}^k \sum_{u=0}^{m_j-1} A_{ju} x^u e^{r_j x} + \sum_{j=1}^h \sum_{u=0}^{\omega_j-1} (B_{ju} \cos(\beta_j x) + C_{ju} \sin(\beta_j x)) x^u e^{\alpha_j x},$$

with parameters A_{ju}, B_{ju}, C_{ju} .

ii Exponential order

For any solution $f(t)$ of any linear nonhomogenous ODE of one variable t , there exists some $M, T \geq 0$ and $\alpha \in \mathbb{R}$ such that for all $t > T$ $|f(t)| \leq e^{M\alpha t}$, and the smallest such α is the largest real part among all roots of its characteristic equation.

iii Example: simple harmonic motion

ODE:

$$\ddot{x} = -\omega^2 x, \quad \omega \in \mathbb{R}_{>0}.$$

Solutions:

$$x = A \sin(\omega t + \phi), \quad A, \phi \in \mathbb{R}.$$

Proof.

$$\ddot{x} + \omega^2 x = 0$$

$$r^2 + \omega^2 = 0$$

$$r = \pm \omega i$$

$$x(t) = A \sin(\omega t) + B \cos(\omega t) = A \sin(\omega t + \phi)$$

□

iv Example: linearly damped simple harmonic motion

ODE:

$$\ddot{x} = -k(x - C) - b\dot{x}, \quad k, b \in \mathbb{R}_{>0} \wedge C \in \mathbb{R}.$$

Solutions:

- Case $b^2 < 4k$ called underdamped:

$$x = A e^{\frac{-b}{2}t} \sin\left(\frac{\sqrt{4k - b^2}}{2}t + \phi\right) + C.$$

- Case $b^2 = 4k$ called critically damped:

$$x = A(1 + Bt)e^{\frac{-b}{2}t} + C$$

- Case $b^2 > 4k$ called overdamped:

$$x = A e^{\frac{-b}{2}t} \left(e^{\frac{\sqrt{b^2 - 4k}}{2}t} + B e^{\frac{-\sqrt{b^2 - 4k}}{2}t} \right) + C$$

Proof.

$$\lambda^2 + b\lambda + k = 0$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4k}}{2}$$

Case $b^2 < 4k$:

$$\lambda = \frac{-b \pm \sqrt{4k - b^2}i}{2}$$

$$x - C = D \left(e^{\frac{-b + \sqrt{4k - b^2}i}{2}t} + E e^{\frac{-b - \sqrt{4k - b^2}i}{2}t} \right) = A e^{\frac{-b}{2}t} \sin \left(\frac{\sqrt{4k - b^2}}{2}t + \phi \right)$$

Case $b^2 = 4k$:

$$\lambda = \frac{-b}{2}$$

$$x - C = A(1 + Bt)e^{\frac{-b}{2}t}$$

Case $b^2 > 4k$:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4k}}{2}$$

$$x - C = A \left(e^{\frac{-b + \sqrt{b^2 - 4k}}{2}t} + B e^{\frac{-b - \sqrt{b^2 - 4k}}{2}t} \right) = A e^{\frac{-b}{2}t} \left(e^{\frac{\sqrt{b^2 - 4k}}{2}t} + B e^{\frac{-\sqrt{b^2 - 4k}}{2}t} \right)$$

□

XI Method of undetermined coefficients (MUC) for constant coefficient linear nonhomogeneous ODE

i Method

Consider an n th-order constant coefficient linear nonhomogeneous ODE

$$\sum_{j=0}^n a_j y^{(j)} = g(x), \quad a_n \neq 0,$$

with $g(x)$ being able to be written in the form

$$g(x) = \sum_{j=1}^p k_j x^{m_j} e^{\alpha_j x} + \sum_{j=p+1}^{p+q} x^{m_j} e^{\alpha_j x} (k_j \cos(\beta_j x) + l_j \sin(\beta_j x)),$$

for some constants $p, q, m_j \in \mathbb{N}_0$ and $\alpha_j, \beta_j, k_j, l_j \in \mathbb{R}$.

The formation of its general solutions using the method of undetermined coefficients is as follows:

1. Solve the complementary equation

$$\sum_{j=0}^n a_j y^{(j)} = 0, \quad a_n \neq 0$$

using characteristic equation to obtain its general solutions y_c .

2. For each term $k_j x^{m_j} e^{\alpha_j x}$ of $g(x)$ for $j \in \mathbb{N} \wedge j \leq p$, construct a particular solution y_j :

(a) If α_j is not a root of the characteristic equation of the complementary equation, let $s_j = 0$; otherwise, let s_j be the multiplicity of α_j as a root of the characteristic equation of the complementary equation.

(b) The ansatz is

$$y_j = x^{s_j} e^{\alpha_j x} \sum_{k=0}^{m_j} C_{jk} x^k,$$

with constants $C_{jk} \in \mathbb{R}$.

(c) Solve for C_{jk} such that y_j is a particular solution of the constant coefficient linear nonhomogeneous ODE

$$\sum_{j=0}^n a_j y^{(j)} = k_j x^{m_j} e^{\alpha_j x}.$$

3. For each term $x^{m_j} e^{\alpha_j x} (k_j \cos(\beta_j x) + l_j \sin(\beta_j x))$ of $g(x)$ for $j \in \mathbb{N} \wedge p < j \leq p+q$, construct a particular solution y_j :

(a) If $\alpha_j \pm \beta_j i$ is not a pair of conjugate imaginary roots of the characteristic equation of the complementary equation, let $s_j = 0$; otherwise, let s_j be the multiplicity of $\alpha_j \pm \beta_j i$ as a pair of conjugate imaginary roots of the characteristic equation of the complementary equation.

(b) The ansatz is

$$y_j = x^{s_j} e^{\alpha_j x} \sum_{k=0}^{m_j} (C_{jk} \cos(\beta_j x) + D_{jk} \sin(\beta_j x)) x^k,$$

with constants $C_{jk}, D_{jk} \in \mathbb{R}$.

(c) Solve for C_{jk}, D_{jk} such that y_j is a particular solution of the constant coefficient linear nonhomogeneous ODE

$$\sum_{j=0}^n a_j y^{(j)} = x^{m_j} e^{\alpha_j x} (k_j \cos(\beta_j x) + l_j \sin(\beta_j x)).$$

4. A particular solution of the origin ODE is

$$y_p = \sum_{j=1}^{p+q} y_j,$$

and the general solutions of the origin ODE is

$$y = y_c + y_p.$$

ii Example: Forced oscillation and resonance

PLACEHOLDER

XII Formation of solutions of linear nonhomogeneous equation

i Wrońskian or Wronskian

For n functions $f_1, f_2, \dots, f_n$ that are $(n-1)$ times differentiable on an interval I , the Wrońskian (also called Wronskian) of them, $W(f_1, f_2, \dots, f_n)$, is a function on I defined by

$$W(f_1, f_2, \dots, f_n)(x) = \det \begin{bmatrix} f_1(x) & f_2(x) & \cdots & f_n(x) \\ f_1'(x) & f_2'(x) & \cdots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x) \end{bmatrix}.$$

If $f_1, f_2, \dots, f_n$ are not linearly independent, then $W = 0$.

ii Second-order Abel's (differential equation) identity/formula

A second-order linear homogeneous ODE

$$y'' + p(x)y' + q(x)y = 0$$

on an interval I with continuous function p and q . The Wrońskian $W(y_1, y_2)$ of two solutions y_1, y_2 of it satisfies

$$W(y_1, y_2)(x) = W(y_1, y_2)(x_0) \cdot \exp\left(-\int_{x_0}^x p(t) dt\right), \quad x \in I,$$

for any $x_0 \in I$.

Proof.

$$\begin{aligned} W' &= (y_1 y_2' - y_1' y_2)' \\ &= y_1' y_2' + y_1 y_2'' - y_1' y_2' - y_1'' y_2 \\ &= y_1 y_2'' - y_1'' y_2 \\ &= y'' = -(p y' + q y). \end{aligned}$$

$$W' = -y_1(p y_2' + q y_2) + y_2(p y_1' + q y_1) = -pW$$

$W' + p(x)W = 0$ is a first-order linear homogenous ODE that can be solved by the integrating factor

$$\mu(x) = \exp\left(\int p(x) dx\right)$$

and initial condition $W(x_0)$:

$$\left(W \exp\left(\int_a^x p(t) dt\right)\right)' = 0$$

$$W \exp\left(\int_a^x p(t) dt\right) = C$$

$$W(x_0) \exp\left(\int_a^{x_0} p(t) dt\right) = C$$

$$W(x) = W(x_0) \cdot \exp\left(\int_a^{x_0} p(t) dt - \int_a^x p(t) dt\right) = W(x_0) \cdot \exp\left(-\int_{x_0}^x p(t) dt\right).$$

□

iii General Abel's (differential equation) identity/formula

A linear homogeneous ODE

$$\sum_{i=0}^{n-1} p_i(x)y^{(i)} + y^{(n)} = 0$$

on an interval I with continuous functions p_i . The Wrońskian $W(y_1, y_2, \dots, y_n)$ of n solutions $y_1, y_2, \dots, y_n$ of it satisfies

$$W(y_1, y_2, \dots, y_n)(x) = W(y_1, y_2, \dots, y_n)(x_0) \cdot \exp\left(-\int_{x_0}^x p_{n-1}(t) dt\right), \quad x \in I,$$

for any $x_0 \in I$.

Proof. Differentiate row by row:

$$\begin{aligned} W' &= \frac{d}{dx} \left(\det \begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix} \right) \\ &= \det \begin{bmatrix} y_1' & y_2' & \cdots & y_n' \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix} + \det \begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1'' & y_2'' & \cdots & y_n'' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix} + \dots + \det \begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n)} & y_2^{(n)} & \cdots & y_n^{(n)} \end{bmatrix} \end{aligned}$$

All these determinants except the last one contain linearly dependent rows and thus equal to 0.

$$W' = \det \begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n)} & y_2^{(n)} & \cdots & y_n^{(n)} \end{bmatrix}.$$

For all $i \in \mathbb{N} \wedge i \leq n$:

$$y_i^{(n)} + \sum_{i=0}^{n-2} p_i(x)y^{(i)} = -p_{n-1}(x)y^{(n-1)}.$$

Hence, we can add to the last row of the above determinant p_i times its $(i+1)$ th row for all $i \in \mathbb{N}_0 \wedge i < n-1$, the value of the determinant is unchanged, and we get

$$W' = \det \begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ -p_{n-1}(x)y_1^{(n-1)} & -p_{n-1}(x)y_2^{(n-1)} & \cdots & -p_{n-1}(x)y_n^{(n-1)} \end{bmatrix} = -p_{n-1}(x)W.$$

$W' + p_{n-1}(x)W = 0$ is a first-order linear homogenous ODE that can be solved by the integrating factor

$$\mu(x) = \exp\left(\int p_{n-1}(x) dx\right)$$

and initial condition $W(x_0)$:

$$\left(W \exp\left(\int_a^x p_{n-1}(t) dt\right) \right)' = 0$$

$$\begin{aligned}
W \exp\left(\int_a^x p_{n-1}(t) dt\right) &= C \\
W(x_0) \exp\left(\int_a^{x_0} p_{n-1}(t) dt\right) &= C \\
W(x) &= W(x_0) \cdot \exp\left(\int_a^{x_0} p_{n-1}(t) dt - \int_a^x p_{n-1}(t) dt\right) = W(x_0) \cdot \exp\left(-\int_{x_0}^x p_{n-1}(t) dt\right).
\end{aligned}$$

□

iv Variation of parameters or variation of constants

Given an n th-order linear nonhomogeneous ODE

$$y^{(n)} + \sum_{i=0}^{n-1} p_i(x)y^{(i)} = q(x),$$

let $y_1(x), y_2(x), \dots, y_n(x)$ be a basis of the vector space of the solutions of the complementary equation,

$$y^{(n)} + \sum_{i=0}^{n-1} p_i(x)y^{(i)} = 0,$$

which can be obtained by variation of parameters of the family of the solutions of the complementary equation. The particular solution of the original ODE (or family of general solutions with the integral constants being parameters) is given by:

$$y_p = \sum_{i=1}^n y_i(x) \int \frac{W_i(x)}{W(x)} dx,$$

where

- $W(x)$ is the Wrońskian of $y_1(x), y_2(x), \dots, y_n(x)$,
- $W_i(x)$ is the Wrońskian of $W(x)$ but with the i th column replaced by $(0, 0, \dots, 0, q(x))$.

For second-order equation

$$y'' + p_1(x)y' + p_0(x)y = q(x),$$

let $y_1(x), y_2(x)$ be two linearly independent solutions of its complementary equation

$$y'' + p_1(x)y' + p_0(x)y = 0,$$

the particular solution (or family of general solutions with the integral constants being parameters) is given by

$$y_p(x) = -y_1(x) \int \frac{y_2(x)q(x)}{y_1(x)y_2'(x) - y_2(x)y_1'(x)} dx + y_2(x) \int \frac{y_1(x)q(x)}{y_1(x)y_2'(x) - y_2(x)y_1'(x)} dx.$$

Proof. The particular solution is given by

$$y_p = \sum_{i=1}^n c_i(x)y_i(x),$$

where $c_i(x)$ are differentiable functions which are assumed to satisfy the $n - 1$ conditions

$$\sum_{i=1}^n c'_i(x) y_i^{(j)}(x) = 0$$

for all $j \in \mathbb{N}_0 \wedge j \leq n - 2$.

By repeated differentiation, we obtain:

$$y_p^{(j)} = \sum_{i=1}^n c_i(x) y_i^{(j)}(x)$$

for all $j \in \mathbb{N}_0 \wedge j \leq n - 1$.

One last differentiation gives

$$y_p^{(n)} = \sum_{i=1}^n c_i(x) y_i^{(n)}(x) + \sum_{i=1}^n c'_i(x) y_i^{(n-1)}(x).$$

Substituting y_p into the original ODE gives:

$$y_p^{(n)} + \sum_{i=0}^{n-1} p_i(x) y_p^{(i)} = q(x),$$

$$\sum_{i=1}^n c_i(x) y_i^{(n)}(x) + \sum_{i=1}^n c'_i(x) y_i^{(n-1)}(x) + \sum_{i=0}^{n-1} p_i(x) \sum_{i=1}^n c_i(x) y_i^{(j)}(x) = q(x),$$

$$\sum_{i=1}^n c'_i(x) y_i^{(n-1)}(x) + \sum_{i=1}^n c_i(x) \left(y_i^{(n)}(x) + \sum_{i=0}^{n-1} p_i(x) y_i^{(j)}(x) \right) = q(x),$$

$$\sum_{i=1}^n c'_i(x) y_i^{(n-1)}(x) = q(x).$$

We obtain a system of linear equations

$$\sum_{i=1}^n c'_i(x) y_i^{(j)}(x) = 0$$

for all $j \in \mathbb{N}_0 \wedge j \leq n - 2$, and

$$\sum_{i=1}^n c'_i(x) y_i^{(n-1)}(x) = q(x).$$

By Cramer Rule, we have

$$c'_i(x) = \frac{W_i(x)}{W(x)}, \quad i \in \mathbb{N} \wedge i \leq n.$$

□

XIII Reduction of order

i Method

Given an n th-order linear homogeneous ODE

$$\sum_{i=0}^n p_i(x)y^{(i)} = 0,$$

if a nontrivial particular solution $y_1(x)$ has been known, we can reduce the order of the ODE by 1 by the substitution

$$y = vy_1$$

and then the substitution

$$u = v'$$

to an $(n - 1)$ th-order linear ODE.

ii Closed-form formula for second-order linear homogeneous ODEs

Given a second-order linear homogeneous ODE

$$y'' + p(x)y' + q(x)y = 0,$$

if a nontrivial particular solution $y_1(x)$ that has been known, then the general solutions of the ODE is

$$y = y_1 \int \frac{e^{-\int p(x)dx}}{y_1(x)^2} dx.$$

Proof. Substitute

$$y = vy_1,$$

$$y' = v'y_1 + vy_1',$$

$$y'' = v''y_1 + 2v'y_1' + vy_1'',$$

$$y_1'' = -py_1' - qy_1,$$

$$y'' = -py' - qy,$$

$$-pv'y_1 - pvy_1' - qvy_1 = v''y_1 + 2v'y_1' - pvy_1' - qvy_1,$$

$$y_1v'' + (2y_1' + py_1)v' = 0,$$

Substitute

$$u = v',$$

$$y_1u' + (2y_1' + py_1)u = 0,$$

$$u' + \left(\frac{2y_1'}{y_1} + p\right)u = 0,$$

$$\mu(x) = \exp\left(\int \frac{y_1'}{y_1} + p dx\right) = y_1(x)^2 e^{\int p(x)dx},$$

$$(u\mu(x))' = 0,$$

$$u = \frac{C}{\mu(x)} = C \frac{e^{-\int p(x)dx}}{y_1(x)^2} = \frac{e^{-\int p(x)dx}}{y_1(x)^2},$$

$$v = \int u dx = \int \frac{e^{-\int p(x)dx}}{y_1(x)^2} dx,$$

$$y = v y_1 = y_1 \int \frac{e^{-\int p(x)dx}}{y_1(x)^2} dx.$$

□

iii Example:

PLACEHOLDER

XIV Power series solutions and Frobenius method for second-order linear homogenous ODE

i Power series solutions

Consider a second-order linear homogenous ODE

$$f(x)y'' + g(x)y' + h(x)y = 0.$$

Let

$$p(x) = \frac{g(x)}{f(x)}, \quad q(x) = \frac{h(x)}{f(x)}.$$

For any a in the intersection of the domain of p and the domain of q , a is called an ordinary point iff p and q are both analytic at a .

1. The power series method seeks a solution in the form:

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n,$$

where x_0 is a fixed point called expansion point and a_n are constants to be determined, called power series.

2. Differentiate the series term by term:

$$y' = \sum_{n=0}^{\infty} a_n n (x - x_0)^{n-1} = \sum_{n=0}^{\infty} a_{n+1} (n+1) (x - x_0)^n,$$

$$y'' = \sum_{n=0}^{\infty} a_n n(n-1) (x - x_0)^{n-2} = \sum_{n=0}^{\infty} a_{n+2} (n+2)(n+1) (x - x_0)^n.$$

3. Substitute into the ODE, rewrite $p(x)$ and $q(x)$ into Taylor expansion at x_0 , rewrite the infinite series products (if any) into Cauchy products, and rewrite the LHS as a power series in x^n .
4. Solve for the recurrence relation of a_n . We may set a_0 as any nonzero constant, typically 1.

5. The two-parameter family of general solutions for all $0 < |x - x_0| < R$ where R is the distance between x_0 and the nearest to x_0 singular point of $p(x)$ or $q(x)$ with parameters a_0, a_1 is:

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n.$$

ii Frobenius method

Consider a second-order linear homogenous ODE

$$f(x)y'' + g(x)y' + h(x)y = 0.$$

Let

$$p(x) = \frac{g(x)}{f(x)}, \quad q(x) = \frac{h(x)}{f(x)}.$$

For any x_0 in the intersection of the domain of p and the domain of q , x_0 is called a regular singular point iff it is not an ordinary point and $(x - x_0)p(x)$ and $(x - x_0)^2q(x)$ are both analytic at a .

1. The Frobenius method seeks a solution in the form:

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}, \quad a_0 \neq 0,$$

where x_0 is called expansion point and a_n are constants to be determined, called Frobenius series.

2. Differentiate the series term by term:

$$y' = \sum_{n=0}^{\infty} a_n (n+r)(x - x_0)^{n+r-1},$$

$$y'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1)(x - x_0)^{n+r-2}.$$

3. Substitute into $(x - x_0)^2 y''$, $(x - x_0)^2 p(x)y'$, and $(x - x_0)^2 q(x)y$ respectively, rewrite $(x - x_0)^2 p(x)$ and $(x - x_0)^2 q(x)$ into Taylor expansion at x_0 , rewrite the infinite series products (if any) into Cauchy products, and let the sum of all terms involving a_0 of the three be L .
4. The indicial polynomial $I(r)$ is defined as the coefficient of the lowest power term of L , and the indicial equation is defined as

$$I(r) = 0,$$

which is a quadratic equation for r .

5. Solve the indicial equation for r . Let the roots be $r_1 \geq r_2$.
6. Depending on the relation of r_1 and r_2 :

- Case $r_1 - r_2 \notin \mathbb{Z} \wedge r_1, r_2 \in \mathbb{R}$: Two linearly independent solutions for all $0 < |x - x_0| < R$ where R is the distance between x_0 and the nearest to x_0 regular singular point of $(x - x_0)p(x)$ or $(x - x_0)^2 q(x)$ are:

$$y_1(x) = (x - x_0)^{r_1} \sum_{n=0}^{\infty} A_n (x - x_0)^n, \quad A_0 \neq 0 \wedge A_n \in \mathbb{R},$$

$$y_2(x) = (x - x_0)^{r_2} \sum_{n=0}^{\infty} B_n (x - x_0)^n, \quad B_0 \neq 0 \wedge B_n \in \mathbb{R}.$$

- Case r_1, r_2 are conjugate imaginary roots: Let $r_1, r_2 = \alpha \pm \beta i$. Two linearly independent solutions for all $0 < |x - x_0| < R$ where R is the distance between x_0 and the nearest to x_0 regular singular point of $(x - x_0)p(x)$ or $(x - x_0)^2 q(x)$ are:

$$y_1(x) = (x - x_0)^\alpha \cos(\beta \ln|x - x_0|) \sum_{n=0}^{\infty} A_n (x - x_0)^n, \quad A_0 \neq 0 \wedge A_n \in \mathbb{R},$$

$$y_2(x) = (x - x_0)^\alpha \sin(\beta \ln|x - x_0|) \sum_{n=0}^{\infty} B_n (x - x_0)^n, \quad B_0 \neq 0 \wedge B_n \in \mathbb{R}.$$

- Case $r_1 - r_2 \in \mathbb{N}_0$: Two solutions for all $0 < |x - x_0| < R$ where R is the distance between x_0 and the nearest to x_0 regular singular point of $(x - x_0)p(x)$ or $(x - x_0)^2 q(x)$ are:

$$y_1(x) = (x - x_0)^{r_1} \sum_{n=0}^{\infty} A_n (x - x_0)^n, \quad A_0 \neq 0 \wedge A_n \in \mathbb{R},$$

$$y_2(x) = (x - x_0)^{r_2} \sum_{n=0}^{\infty} B_n (x - x_0)^n, \quad B_0 \neq 0 \wedge B_n \in \mathbb{R}.$$

If they both exist and are linearly independent, no modification is needed; otherwise, y_2 is modified to

$$y_2(x) = \ln|x - x_0| y_1(x) + (x - x_0)^{r_2} \sum_{n=0}^{\infty} B_n (x - x_0)^n, \quad B_0 \neq 0 \wedge B_n \in \mathbb{R}.$$

7. Substitute y_1, y_1', y_1'' and y_2, y_2', y_2'' into the ODE respectively, rewrite $p(x)$ and $q(x)$ into Taylor expansion at x_0 , rewrite the products into Cauchy products, and rewrite LHSs as power series in x^n .

8. Solve for the recurrence relations of A_n and B_n .

- Case $r_1 - r_2 \notin \mathbb{Z} \wedge r_1, r_2 \in \mathbb{R}$ and Case r_1, r_2 are conjugate imaginary roots: We may set A_0 and B_0 as any nonzero constants (typically 1).
- Case $r_1 - r_2 \in \mathbb{N}_0$: The terms multiplied by $\ln|x - x_0|$ in the substitution of $y_2(x)$ yield the same recurrence relation for A_n as that obtained from the substitution of $y_1(x)$, and therefore need not be solved again. We may set A_0 as any nonzero constant (typically 1). The lowest-two-order equations in the substitution of $y_2(x)$ may determine B_0 and/or B_1 , or relation of B_0 and B_1 . If one of them remains free after applying this constraint, it may be chosen arbitrarily except that $B_0 \neq 0$, with the remaining determined uniquely by the recurrence relation.

9. The two-parameter family of general solutions for all $0 < |x - x_0| < R$ where R is the distance between x_0 and the nearest to x_0 regular singular point of $(x - x_0)p(x)$ or $(x - x_0)^2q(x)$ is:

$$y = C_1 y_1(x) + C_2 y_2(x), \quad C_1, C_2 \in \mathbb{R}.$$

Proof. PLACEHOLDER □

iii Example: $y'' - 2xy' + y = 0$

ODE:

$$y'' - 2xy' + y = 0.$$

Solutions:

$$y = \sum_{n=0}^{\infty} a_n x^n,$$

$$a_0, a_1 \in \mathbb{R}, \quad a_{n+2} = \frac{n-1}{(n+2)(n+1)} a_n.$$

Proof.

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$y' = \sum_{n=0}^{\infty} a_{n+1} (n+1) x^n$$

$$y'' = \sum_{n=0}^{\infty} a_{n+2} (n+2)(n+1) x^n$$

$$a_{n+2} (n+2)(n+1) - a_n n + a_n = 0$$

$$a_{n+2} = \frac{n-1}{(n+2)(n+1)} a_n$$

□

iv Example: $y'' + \sin(x)y' + x^2 y = 0$

ODE:

$$y'' + \sin(x)y' + x^2 y = 0.$$

Solutions:

$$y = \sum_{n=0}^{\infty} a_n x^n,$$

where:

$$a_{-1} = a_{-2} = 0, \quad a_0, a_1 \in \mathbb{R},$$

$$a_{n+2} (n+2)(n+1) + \sum_{k=0}^{\left\lfloor \frac{n-1}{2} \right\rfloor} \frac{(-1)^k}{(2k+1)!} a_{n-2k} (n-2k) + a_{n-2} = 0.$$